

MAT 4123/6973
History of Mathematics
Spring 2024

Instructor: Sandy Norman
SB 1.xx.xx

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210-422-2284

Office Hours: MW 6:00 p.m. – 7:00 p.m. and by appt.
or via Zoom <https://utsa.zoom.us/j/8522459681>

Class webpage: xxxxxx

Class meeting: MWF 7:30-8:45 p.m. MS 2.03.08

Overview. During this course we will take a broad view of the history of mathematics from prehistoric times to the present, with some additional attention to important figures in the development or dissemination of mathematical ideas, critical points in the evolution (sometimes revolution) of mathematics, and a closer look at some of the interesting aspects of mathematics, both ancient and modern.

Prerequisites. For MAT 4123 the prereqs are (MAT 3123 or MAT 4263) and (MAT 3233 or MAT 4233) or permission of instructor. For MAT 6973 the prereq is consent of instructor.

Text/Resources: The required text for the course (*The Mathematical Experience* by Davis and Hersh) will be provided. There will also be required readings provided from various sources. Furthermore, any of the following resources might prove useful:

A History of Mathematics – Merzbach & Boyer
Mathematics and Its History – Stillwell (Springer)
The Story of Proof: Logic and the History of Mathematics – Stillwell
History of Mathematical Notations -- Cajori

Grading. Regular in-class presentations (including problem presentations from occasional homework assignments), an exam, and calendar reports, will comprise the course work for the class.

Presentations	60 pts
Exam	20 pts
Reports	20 pts

Letter grades will be determined based on the following table:

A+ [99, 100]	B+ [88, 89)	C+ [78, 79)	D+ [68, 69)	
A [90, 99)	B [80, 88)	C [70, 78)	D [60, 69)	F [0, 59)
A- [89,90)	B- [79, 80)	C- [69, 70)	D- [59, 60)	

Outline. A *tentative* outline for the course is included below.

Week	Topic
1	Dawn of mathematics
2	Early mathematics around the globe
3	Number systems and arithmetic
4	Pythagoreans and the rise of irrational numbers
5	Euclid, geometry, proof, and axiomatic systems
6	Fibonacci and the mathematical renaissance
7	Algebra and Descartes
8	Calculus from Archimedes to Newton
9	A revolution in geometry
10	Analysis
11	Statistical mechanics and mathematical physics
12	Abstract algebra
13	Cantor and set theory
14	Rise of data science

Ombudsperson: If you need to discuss any problems or issues, please contact Ilse Rosales Cooper at ilsa.rosalesmartinez@utsa.edu.

[Student Guide to Online Learning](#)

This is a guide produced by the University to help students deal with and manage their online learning experience. You might find it helpful, if you are registered in any online courses or have struggled with online courses over this past few semesters.

Additional information about Counseling Services, Student with Disabilities, Tutoring Services, and other relevant topics can be found at

Other information required for syllabi can be found at <http://provost.utsa.edu/syllabus.asp> or [Course Syllabi Instructions](#), [Common Syllabus Information](#), and [Inclusivity Statement](#).

Public Health Considerations: The health and safety of our campus community is a shared responsibility of all Roadrunners. It is important to note that none of us can guarantee a COVID-19 free environment. We all must, however, follow the guidelines outlined in the [UTSA Public Health Task Force Report](#) (“Report”) and any other

applicable policies as may be communicated by the University from time to time. This will include regulating behaviors outlined in the Report including:

- Encouraging the use of [face coverings](#),
- Self-monitoring for symptoms using the [Daily Health Check](#) before coming to campus,
- Getting tested for COVID-19 if showing symptoms or after a close contact with a COVID-19 positive individual (if you are not already fully vaccinated and are not symptomatic),
- Following proper hygiene practices, including frequent hand sanitization, using cleansing wipes to disinfect surfaces, and minimizing the use of shared devices, tools and equipment,
- Avoid congregating (i.e. bottle-necking) near the entrances and exits before and after class – keeping your distance to reduce possible transmission from symptomatic or asymptomatic individuals.
- Communicating any COVID-19 related health concern to your supervisor or professor, and
- [Submitting a self-report](#) to report your positive test results or exposure (if not fully vaccinated and are also symptomatic for COVID-19 infection).

In turn, faculty members or supervisors will submit a COVID Case Referral to alert the COVID Response Team about positive COVID-19 cases for operational action. Failure to abide by these guidelines and requirements may result in disciplinary action in accordance with the [Student Code of Conduct](#) or applicable employment policies and procedures. Violations should be reported to the Office of Institutional Compliance via the [UTSA Hotline](#) for appropriate action.

For Face-to-Face Classroom and Other Academic Sessions: Face coverings/masks are recommended in indoor public and common spaces, especially for those individuals who are high risk and/or not vaccinated. For the latest information, please review the [Roadrunner Roadmap](#).

Common Syllabus Information: The latest version of the following text is available at: <https://provost.utsa.edu/syllabus.html>

Counseling Services: Counseling Services provides confidential, professional services by staff psychologists, social workers, counselors and psychiatrists to help meet the personal and developmental needs of currently enrolled students. Services include individual brief therapy for personal and educational concerns, couples/relationship counseling, and group therapy on topics such as college adaptation, relationship concerns, sexual orientation, depression and anxiety. Counseling Services also screens for possible learning disabilities and has limited psychiatric services. Visit [Counseling Services](#) or call (210) 458- 4140 (Main Campus) or (210) 458-2930 (Downtown Campus).

Student Code of Conduct and Scholastic Dishonesty: The Student Code of Conduct is Section B of the Appendices in the Student Information Bulletin. Scholastic Dishonesty is listed in the [Student Code of Conduct](#) (Sec. B of the Appendices) under Sec. 203.

Students with Disabilities: The University of Texas at San Antonio in

compliance with the Americans with Disabilities Act and Section 504 of the Rehabilitation Act provides “reasonable accommodations” to students with disabilities. Only those students who have officially registered with Student Disability Services and requested accommodations for this course will be eligible for disability accommodations. Instructors at UTSA must be provided an official notification of accommodation through Student Disability Services. Information regarding diagnostic criteria and policies for obtaining disability-based academic accommodations can be found at www.utsa.edu/disability or by calling Student Disability Services at (210) 458-4157. Accommodations are not retroactive.

Transitory/Minor Medical Issues: In situations where a student experiences a transitory/minor medical condition (e.g. broken limb, acute illness, minor surgery) that impacts their ability to attend classes, access classes or perform tasks within the classroom over a limited period of time, the student should refer to the class attendance policy in their syllabus.

Supplemental Instruction: Supplemental Instruction offers student-led study groups using collaborative learning for historically difficult classes. Supported courses and schedules can be found on the [TRC website](#). You can call the SI office if you have questions or for more information at (210) 458-7251.

Tutoring Services: Tomás Rivera Center (TRC) may assist in building study skills and tutoring in course content. The TRC has several locations at the Main Campus and is also located at the Downtown Campus. For more information, visit the [Tutoring Services web page](#) or call (210) 458-4694 on the Main Campus or (210) 458-2838 on the Downtown Campus.

Academic Success Coaching: The Tomás Rivera Center (TRC) Academic Success Coaching Program offers one-on-one study skills assistance through Academic Coaching. Students meet by appointment with a professional to develop more effective study strategies and techniques that can be used across courses. Group workshops are also offered each semester to help students defeat common academic challenges. Find out more information on the TRC Academic Success Coaching [website](#) or call (210) 458-4694.

Sexual Harassment and Sexual Misconduct: UTSA is committed to providing an environment free from all forms of discrimination and sexual harassment, including sexual misconduct, sexual assault, domestic violence, dating violence, and stalking. If a student has experienced or experiences any of these incidents, know that UTSA has resources to help.

UTSA faculty have the responsibility to create a learning environment that is safe and free from hostility. State and federal law as well as UTSA’s Handbook of Operating Procedures ([HOP 9.24](#)) require that instructors must report incidents of sexual harassment and sexual misconduct they learn about to the Title IX Coordinator or a Deputy Title IX Coordinator. This means that if a student tells their instructor about a situation (including classroom discussions, written work and/or one-on-one meetings) involving sexual harassment, sexual assault, dating violence, domestic violence, or stalking, the instructor must report it to the [EOS/Title IX Office](#). Although the faculty member must report the situation, the student will still have options about how their case will be handled, including whether or not they wish to pursue a formal complaint. The university’s goal is to

make sure students are aware of the range of options available to them and have access to the resources they need. If a student wishes to speak to someone confidentially, they can contact any of the following on-campus resources, who are not required to report the incident to the EOS/Title IX Office:

(1) [Counseling Services](#) at 210-458-4140; (2) [Student Health Services](#) at 210-458-4142; or (3) [PEACE Center](#) at 210-458-4077.

Campus Safety & Emergency Preparedness: UTSA is committed to providing a safe campus environment for students, faculty, staff, and visitors. As members of the community, we encourage you to take the following actions to be better prepared in case of an emergency:

- Alerts: Ensure you are signed up for UTSA Alerts through your ASAP.utsa.edu account. Emergency Procedures: Read through the emergency response guide on the UTSA Alerts website (www.alerts.utsa.edu)
- Safety App: Download the LiveSafe App on your phone through the Apple store or Google Play; visit the UTSA Alerts website for details.
- Important Numbers: UTSA Police - Emergency: (210) 458-4911; Non-Emergency: (210) 458-4242

Each one of us play a critical role in making sure ALL ROADRUNNERS are safe, know what to do, and how to stay informed during a campus crisis. Don't be scared, be prepared! #UTSAprepared

Inclusivity Statement: The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to ending generations of discrimination and inequity. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery and innovation that embraces the uniqueness of each voice. Learn more at <https://www.utsa.edu/inclusiveexcellence>.

The Roadrunner Creed: The Roadrunner Creed: The University of Texas at San Antonio is a community of scholars, where integrity, excellence, inclusiveness, respect, collaboration, and innovation are fostered.

As a Roadrunner, I will:

- Uphold the highest standards of academic and personal integrity by practicing and expecting fair and ethical conduct;
- Respect and accept individual differences, recognizing the inherent dignity of each person;
- Contribute to campus life and the larger community through my active engagement;
- and Support the fearless exploration of dreams and ideas in the advancement of ingenuity, creativity, and discovery.

Guided by these principles now and forever, I am a Roadrunner!

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